Nurture: Maternal Health Intelligence & Decision Support System

B.Tech (AIDS)

UG-II (III Semester) · Introduction to AI & ML

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Problem Statement

A pregnant woman's record is split across **visits, lab reports, prescriptions and diet plans**; no one sees the whole picture at the visit.

Doctors have minutes per patient to spot danger signs, check drug safety and plan the next tests.

Patients cannot easily see their own reports, medicines or when to come back.

Goal: one system that predicts risk early, checks every prescription, and shows each person — patient, doctor, dietician — what they need.

13.4 M

babies born preterm worldwide in 2020 [1]

3.02 M

preterm births in India — the most of any country [1]

97

maternal deaths per 1,00,000 live births in India, 2018–20 [2]

Literature Review

Paper / System	Method	Result	Limitation
Ahmed et al., 2020 [4] — UCI Maternal Health Risk	IoT vitals (BP, glucose, temp., heart rate) → 3-level risk class	Widely used benchmark	One reading per row, 6 features: no change across visits
Sharifi-Heris et al., 2022 [5] — review of 13 studies	ML on health records to predict preterm birth	ROC-AUC 0.62–0.98, mostly 0.65–0.85	Complex models rarely beat logistic regression; little external validation
Artzi et al., 2020 [6] — Nature Medicine	Gradient boosting on nationwide EHR to predict GDM	ROC-AUC 0.85 (9-question model: 0.80)	Prediction only: no prescribing safety, no patient or dietician view

Gap we address: longitudinal, visit-by-visit prediction without data leakage, combined with rule-based drug safety and role-based views for patient, doctor and dietician.

Objectives of the Project

Clean, trustworthy data

Audit the six linked datasets and fix generator and preprocessing defects.

Early risk prediction

Predict preterm birth, NICU admission and gestational diabetes at each visit, using only data known at that visit.

Honest evaluation

Validate by patient, compare with the best score the data allows, explain every score.

Safe prescribing

Check every medicine against allergies, interactions and pregnancy safety rules.

Doctor in control

Doctor confirms or overrides each risk with a reason; all decisions are logged.

Fast role-based app

Patient, doctor and dietician views with reports, diet plans and appointment booking.

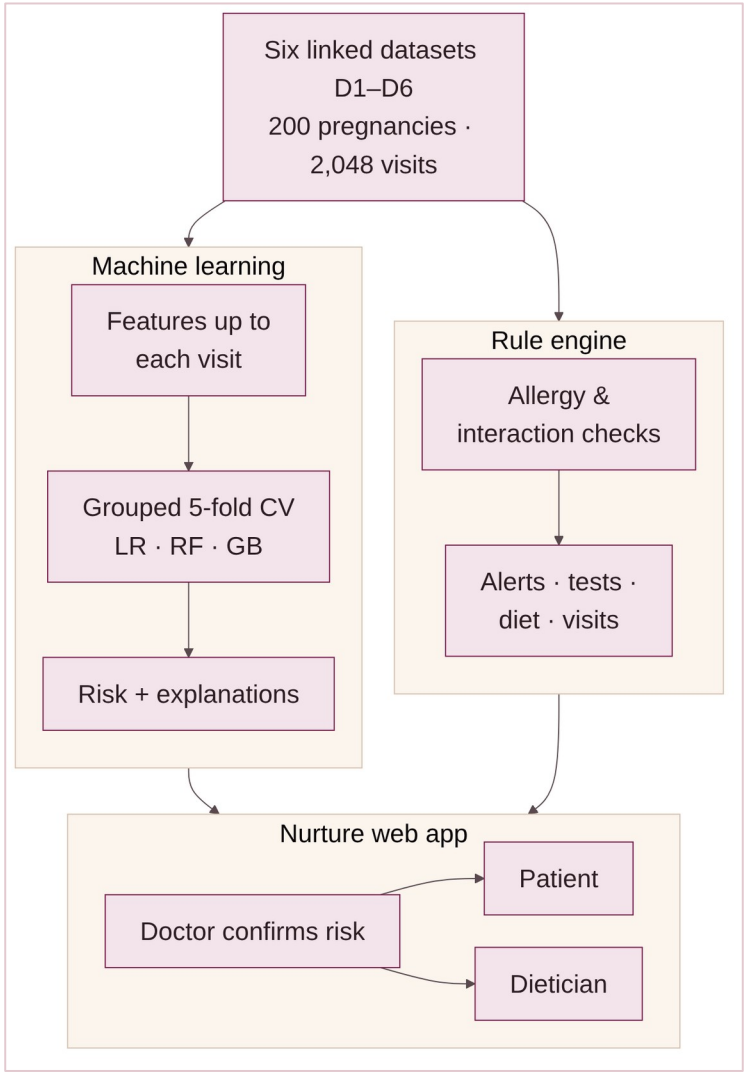
Proposed Methodology

Step 1 – Data collection: six linked synthetic datasets (D1–D6) for 200 Indian pregnancies, grounded in MoHFW, ICMR-NIN and NLEM 2022 guidance.

Step 2 – Preprocessing: fixed 7 data defects; built 49 features per visit from data known up to that visit (no future information).

Step 3 – Model / system development: Logistic Regression, Random Forest and Gradient Boosting; rule engine for allergies, interactions, alerts and test schedule; FastAPI + JavaScript web app.

Step 4 – Testing and evaluation: 5-fold cross-validation grouped by pregnancy; ROC-AUC, balanced accuracy, sensitivity, specificity; ceiling check against the generator's true probabilities.



Tools & Technologies

Area	Tools used
Language	Python 3.12, JavaScript (ES2020)
Data & ML	pandas, NumPy, scikit-learn (Logistic Regression, Random Forest, Gradient Boosting), joblib
Back end	FastAPI web server with Uvicorn; JSON API with gzip compression
Front end	HTML, CSS and plain JavaScript; hand-drawn SVG charts; no framework for speed
Analysis & figures	Matplotlib, Mermaid diagrams, Playwright for automated screenshots and testing
Deployment	Docker image; free hosting on Hugging Face Spaces or Render
Hardware	Standard laptop CPU — no GPU or sensors needed; ~300 MB memory at run time

Experimental Setup

Dataset	One row is	Rows × columns
D1 Patient & pregnancy	one pregnancy	200 × 48
D2 ANC visits	one visit	2,048 × 39
D3 Laboratory	one test (29 types)	12,863 × 19
D4 Medication	one medicine event	874 × 34
D5 Nutrition	one assessment	2,048 × 36
D6 Outcomes	one delivery	200 × 22

Source: synthetic cohort generated for this project; linked by Pregnancy_ID and Visit_ID. Reference tables: 32 medicines, 13 interaction rules, 60 Indian foods.

Samples

2,048 visits from 200 pregnancies

Features

49 per visit: 19 numeric, 2 ordinal, 28 yes/no

Targets (2 classes each)

Preterm birth 35/200 · NICU 14/200 · GDM 55/200

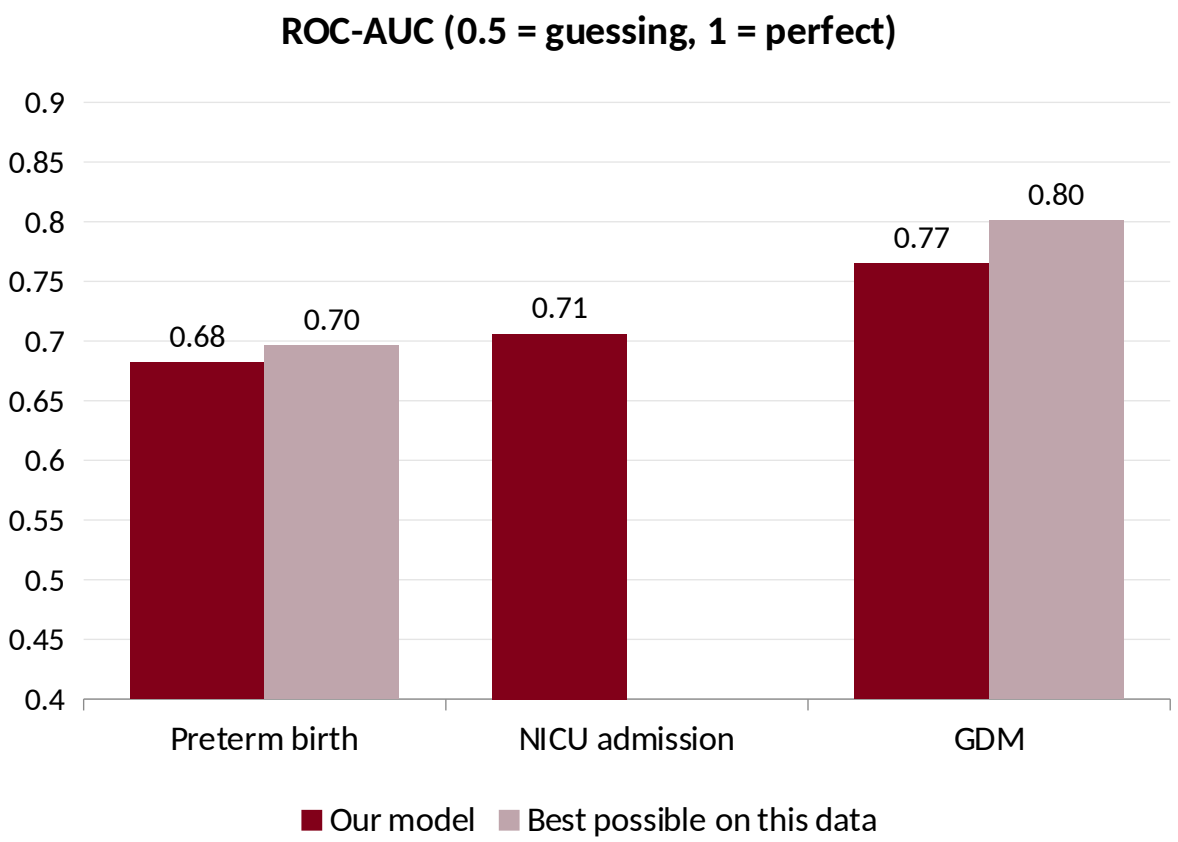
Validation

5-fold cross-validation grouped by pregnancy

Environment

Python 3.12, scikit-learn 1.9, laptop CPU

Results: Model Performance

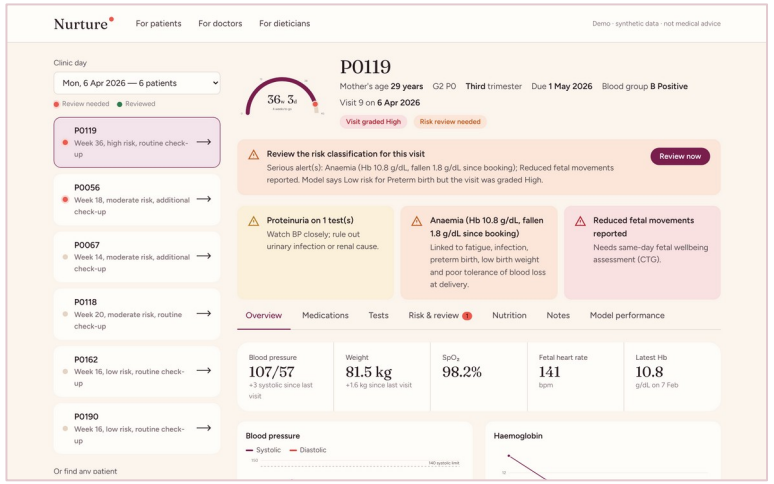


RF = Random Forest, LR = Logistic Regression. Out-of-fold, grouped by pregnancy. NICU has no closed-form ceiling.

Metric	Preterm	NICU	GDM
Best model	RF	RF	LR
Accuracy	0.63	0.72	**0.81**
Always-“no” accuracy	0.83	0.93	0.73
Balanced accuracy	0.65	0.72	0.74
Sensitivity	0.69	0.71	0.60
Specificity	0.62	0.72	0.88

Why accuracy looks low: only 7–18% of women have these outcomes, so always saying “no” scores 83–93%. Our models catch ~70% of real cases and sit within 0.04 of the best ROC-AUC this data allows.

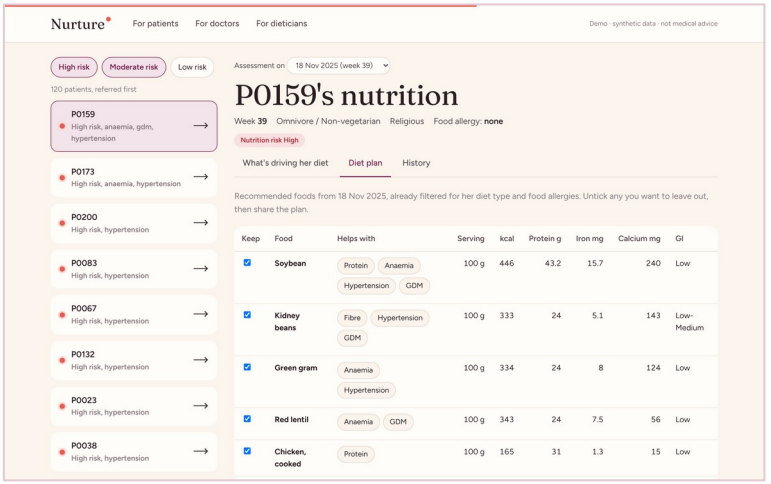
Results: Working Prototype (Nurture web app)



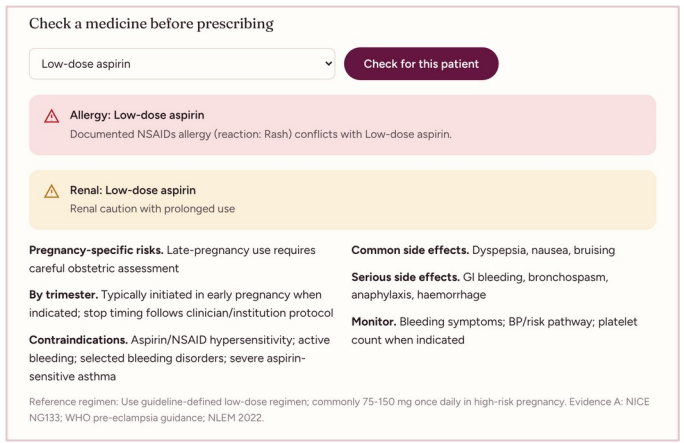
Doctor: clinic list, alerts, risk review



Patient: reports by test with trend



Dietician: worklist and diet plan



Safety check in action: an NSAID allergy conflicts with low-dose aspirin.

The rule engine found 2 allergy conflicts that the original records marked as “passed”.

Speed: 10–130 ms per screen, 70–100 ms to switch patients.

Challenges

Dataset defects

Every pregnancy was labelled GDM; preterm births were inflated to 31%; lab values were scrambled by encoding.

Solved: Fixed 7 defects in generators and preprocessing; preterm now 17.5%, GDM 27.5%.

Low accuracy

Preterm and NICU outcomes are mostly random in the generator, so no model can score highly.

Solved: Computed the best possible score (0.696) and reported balanced metrics honestly.

Data leakage risk

Visit records grow over time; using later visits would fake high accuracy.

Solved: Features use only data up to each visit; validation is grouped by pregnancy.

Slow first interface

The first Streamlit version reloaded the whole page on every click.

Solved: Rebuilt as FastAPI + JavaScript with pre-computed scores: 10–130 ms per screen.

Conclusions & Future Scope

Work completed

Cleaned and regenerated the six-dataset cohort (7 defects fixed).

Three leakage-free models: preterm 0.68, NICU 0.71, GDM 0.77 ROC-AUC — near the data's ceiling.

Rule engine for allergies, interactions, alerts, tests and diet.

Doctor confirms every risk; working web app with booking, ready to deploy.

Impact: risks flagged early, unsafe prescriptions caught, and each person sees what they need.

Before final evaluation

Rewrite outcome generators with effect sizes from published studies.

Calibrate predicted probabilities; test on de-identified real records with ethics approval.

Add login and a database instead of CSV files.

Patient view in Hindi and Kannada with SMS appointment reminders.

Deploy online on Hugging Face Spaces for the demo.

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THANK YOU